Robert Koch

Robert Koch (1843–1910) was a German scientist who won the 1905 Nobel Prize for Medicine. He is regarded as the co-founder (with Louis Pasteur) of modern microbiology. Koch is regarded as one of the greatest bacteriologists the world has ever known. He had a wide range of interests, especially bacteriology, a science that was then in its early stages. Koch's first major finding in bacteriology was in 1876, when he demonstrated that if viable *Bacillus anthracis* was injected into healthy mice, the mice eventually developed anthrax.

While many of the causes of disease are now recognised, in the past people did not realise (or refused to believe) that tiny organisms could be responsible for certain illnesses. Koch was the first person to show conclusively that the cause of an infectious disease was a specific organism. He was also able to identify many of the organisms associated with particular diseases.

1 Define *viable*.

2 Suggest why people may not have believed that microorganisms were responsible for disease.

Koch was one of the first scientists to obtain microbes from infected animals and show how these microbes could be grown in artificial conditions. The techniques he developed were used to isolate many different types of bacteria. He also demonstrated the importance of being able to destroy the cultured microbes safely after research had been carried out. Koch was one of the first people to advocate the sterilising of surgical instruments with steam. Louis Pasteur also promoted the importance of sterilising surgical instruments, bedding and other items used in hospitals.

3 Define *sterilising*.

4 Give an example of when bedding might need to be sterilised.

5 The terms *sterilisation* and *disinfection* are often interchanged. While they are similar in many respects, there is a fine difference between them. Explain the difference between the two terms.

6 Suggest the conditions that would need to be met to grow microbes artificially.

7 Explain why we might want to grow microbes, even if they are harmful.

Tuberculosis (TB) is an infectious disease of the lungs spread by airborne droplets from the sputum of infected people. Estimates vary, but between a quarter and a third of all people have the bacteria causing TB present in their body. The disease tends to be most common where there is overcrowding, poor nutrition and poor health care. It currently kills over two million people a year. In Koch's time it was almost invariably fatal. The treatment was for the patient to be cared for in a sanatorium until they died. Koch worked for many years on isolating and cultivating bacteria. He was highly regarded by his contemporaries. One was so impressed by Koch's work on the bacteria causing anthrax that he stated "This young Robert Koch will surprise and shame us by the brilliance of his investigations". In 1882 Koch isolated the bacteria (Mycobacterium tuberculosis) causing the disease. He developed the tuberculin test in 1890 which helped doctors better diagnose the disease.

Eventually, in 1924, the BCG vaccine for the disease was developed by other doctors. In more recent times antibiotics have been used to treat tuberculosis but the disease has developed a resistance to some of these. It has also become a problem for people whose immune system is not at full strength. One survey in the USA reported that people working in nursing homes are three times more likely to develop TB than similar individuals with different occupations. Domesticated animals such as cattle, chickens and pigs can carry the disease but it is rare in wild animals.

X-rays can show development of TB in the chest but more specific tests are needed to confirm the disease.

8 What is a sanatorium?

9 What is meant by the phrase: 'It was almost invariably fatal'?

10 Is having the presence of TB bacteria present in your body a definite indication that you will catch the disease?

11 Give a reason why more people working in nursing homes might exhibit symptoms of the disease than is generally expected.

12 Give two techniques that can be used to screen for TB.

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Research

13 Find out what the initials *BCG* represent.

14 Summarise the symptoms of tuberculosis. Include a description of how it is spread.

Koch's postulates

In 1890 Robert Koch set out his criteria for judging whether a given bacterium was the cause of a given disease.

- The same bacteria must be found in all cases of a given disease.
- The bacteria must be isolated from the diseased host and grown in pure culture.
- The isolated bacteria must reproduce the same disease when inoculated into a healthy susceptible animal.
- The bacteria must be recoverable from the experimentally infected host.

Limitations of Koch's postulates

- A bacterium that is a normal part of the environment may become a pathogen in certain situations. They can mutate and acquire extra virulence factors, making them pathogenic. Some usually harmless bacteria can cause disease if they gain access to deep tissues by trauma or surgery.
- In immunocompromised patients, many free-living bacteria and components of the normal flora can cause disease, especially if they reach deep tissues.
- Not all of those infected or colonised by a pathogenic bacterium will develop disease. Subclinical infection is usually more common than clinically obvious infection.
- Some bacteria are not culturable in vitro (outside the body and in an artificial environment).

15 What does it mean when a person is *immunocompromised*?

16 Organisms grown *in vivo* are grown inside another organism. Describe how organisms are grown *in vitro*.

17 Explain what is meant when an animal is *susceptible* to a disease.

18 What conditions might cause an animal be more likely to catch a disease and why?

19 Suggest how a mutated form of a usually harmless microorganism might suddenly become dangerous.

20 Discuss why, if Koch's postulates have limitations, they are still used by medical researchers.

21 Suggest why some species of bacteria cannot be cultured in vitro.

Research

22 Koch is famous for his work with five of the diseases listed below. Identify the five diseases.

Anthrax, cholera, Ebola, leukaemia, malaria, sleeping sickness, syphilis, tuberculosis

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